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Method and apparatus having a single coil with embedded magnets

Abstract

Embodiments of the present disclosure include a method and apparatus. An exemplary apparatus includes a solenoid assembly that includes a ferromagnetic core tube having a first channel and a second channel disposed on the radial exterior surface of the ferromagnetic core tube. The first channel and the second channel circumscribe the ferromagnetic core tube along a given portion of the longitudinal axis. The first channel and the second channel are spaced from one another along the longitudinal axis. The first channel has a first magnetic ring radially spaced apart from a first thin wall. The second channel has a second magnetic ring radially spaced part from a second thin wall. The solenoid assembly also includes a coil located radially outward of the ferromagnetic core tube.

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Background/Summary

BACKGROUND OF THE INVENTION

Field of the Invention

(1) The present disclosure relates to a solenoid assembly. The present invention relates more particularly to an improved solenoid assembly having embedded magnets.

Description of Related Art

(2) A solenoid is a type of electromagnet that can include a coil that operable to form magnetic fields. In operation, solenoids typically allow a current to pass through their coil thereby creating a magnetic field within the immediate vicinity of the coil. Solenoids can be used in valves that are integrated devices having an electromechanical solenoid operable to actuate a valve.

BRIEF SUMMARY OF THE INVENTION

(3) In view of the foregoing, it is an object of the present disclosure to provide a method and apparatus.

(4) A first exemplary embodiment of the present disclosure provides a solenoid assembly. The solenoid assembly having a ferromagnetic core tube having a longitudinal axis, a radial exterior surface and a radial interior surface surrounding a cavity, the ferromagnetic core tube comprising a first channel and a second channel disposed on the radial exterior surface of the ferromagnetic core tube, the first channel and the second channel circumscribing the ferromagnetic core tube along a given portion of the longitudinal axis, wherein the first channel and the second channel are spaced from one another along the longitudinal axis, the first channel comprising a plurality of radial exterior surfaces, the second channel comprising a second plurality of radial exterior surfaces, wherein a first portion of the ferromagnetic core tube between the plurality of radial exterior surfaces of the first channel and the radial interior surface of the ferromagnetic core tube define a first thin wall portion of the ferromagnetic core tube, wherein a second portion of the ferromagnetic core tube between the second plurality of radial exterior surfaces of the second channel and the radial interior surface of the ferromagnetic core tube define a second thin wall portion of the ferromagnetic core tube, the first channel comprising a first magnetic ring radially spaced apart from the first thin wall, the second channel comprising a second magnetic ring radially spaced part from second thin wall. The solenoid assembly further includes a coil located radially outward of the ferromagnetic core tube, wherein the ferromagnetic core tube comprises a uniform single piece material.

(5) A second exemplary embodiment of the present disclosure provides a solenoid assembly having a ferromagnetic core tube having a longitudinal axis, a radial exterior surface and a radial interior surface surrounding a cavity, the ferromagnetic core tube comprising a first channel and a second channel disposed on the radial exterior surface of the ferromagnetic core tube, the first channel and the second channel circumscribing the ferromagnetic core tube along a given portion of the longitudinal axis, wherein the first channel and the second channel are spaced from one another along the longitudinal axis, the first channel comprising a plurality of radial exterior surfaces, the second channel comprising a second plurality of radial exterior surfaces, wherein a first portion of the ferromagnetic core tube between the plurality of radial exterior surfaces of the first channel and the radial interior surface of the ferromagnetic core tube define a first thin wall portion of the ferromagnetic core tube, wherein a second portion of the ferromagnetic core tube between the second plurality of radial exterior surfaces of the second channel and the radial interior surface of the ferromagnetic core tube define a second thin wall portion of the ferromagnetic core tube, the first channel comprising a first magnetic ring radially spaced apart from the first thin wall, the second channel comprising a second magnetic ring radially spaced part from second thin wall. The solenoid assembly further includes an armature maintained with the cavity, the armature operable to move through the longitudinal axis of the cavity in response to a current passing through the coil. The solenoid assembly still further includes a coil located radially outward of the ferromagnetic core tube, wherein the ferromagnetic core tube comprises a uniform single piece material, wherein the first channel maintains a filler material between an inner radial surface of the first magnetic ring and the outer radial surface of the first thin wall, and wherein the second channel maintains a filler material between an inner radial surface of the second magnetic ring and the outer radial surface of the second thin wall.

(6) A third exemplary embodiment of the present disclosure provides a method. The method

includes providing a ferromagnetic core tube having a longitudinal axis, a radial exterior surface and a radial interior surface surrounding a cavity, the ferromagnetic core tube comprising a first channel and a second channel disposed on the radial exterior surface of the ferromagnetic core tube, the first channel and the second channel circumscribing the ferromagnetic core tube along a given portion of the longitudinal axis, wherein the first channel and the second channel are spaced from one another along the longitudinal axis, the first channel comprising a plurality of radial exterior surfaces, the second channel comprising a second plurality of radial exterior surfaces, wherein a first portion of the ferromagnetic core tube between the plurality of radial exterior surfaces of the first channel and the radial interior surface of the ferromagnetic core tube define a first thin wall portion of the ferromagnetic core tube, wherein a second portion of the ferromagnetic core tube between the second plurality of radial exterior surfaces of the second channel and the radial interior surface of the ferromagnetic core tube define a second thin wall portion of the ferromagnetic core tube, the first channel comprising a first magnetic ring radially spaced apart from the first thin wall, the second channel comprising a second magnetic ring radially spaced part from second thin wall. The method further includes providing a coil located radially outward of the ferromagnetic core tube, wherein the ferromagnetic core tube comprises a uniform single piece material.

(7) The following will describe embodiments of the present disclosure, but it should be appreciated that the present disclosure is not limited to the described embodiments and various modifications of the disclosure are possible without departing from the basic principles. The scope of the present disclosure is therefore to be determined solely by the appended claims.

Description

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWING(S)

- (1) FIG. 1 presents a cross-sectional view of an exemplary device suitable for use in practicing exemplary embodiments of this disclosure.
- (2) FIG. 2 presents a cross-sectional view of another exemplary device suitable for use in practicing exemplary embodiments of this disclosure.
- (3) FIG. 3 presents yet another cross-sectional view of an exemplary device suitable for use in practicing exemplary embodiments of this disclosure.
- (4) FIG. 4 presents a perspective view of a portion of an exemplary magnetic ring suitable for use in practicing exemplary embodiments of this disclosure.
- (5) FIG. 5 presents a cross-sectional view of a portion of an exemplary magnetic ring suitable for use in practicing exemplary embodiments of this disclosure.
- (6) FIG. 6 presents a close-up cross-sectional view of a portion of an exemplary device suitable for use in practicing exemplary embodiments of this disclosure.
- (7) FIG. 7 presents a cross-sectional view of yet another embodiment of a device suitable for use in practicing exemplary embodiments of this disclosure.
- (8) FIG. 8 presents a close-up cross-sectional view of an exemplary channel on a device suitable for use in practicing exemplary embodiments of this disclosure.
- (9) FIG. 9 presents a close-up cross-sectional view of another exemplary channel on a device suitable for use in practicing exemplary embodiments of this disclosure.
- (10) FIG. 10 presents a close-up cross-sectional view of yet another exemplary channel on a device suitable for use in practicing exemplary embodiments of this disclosure.
- (11) FIG. 11 presents a close-up cross-sectional view of another alternative exemplary channel on a device suitable for use in practicing exemplary embodiments of this disclosure.
- (12) FIG. 12 presents a logic flow diagram in accordance with an exemplary method or process suitable for practicing exemplary embodiments of this disclosure.

(13) FIG. 13 presents a cross-sectional view of an exemplary device with spaced apart magnets suitable for practicing exemplary embodiments of this disclosure.

(14) FIG. 14 presents a cross-sectional view of an exemplary device having a channel spacer suitable for practicing exemplary embodiments of this disclosure.

DETAILED DESCRIPTION OF THE INVENTION

(15) Embodiments of the present disclosure provide a solenoid assembly having a core tube that is circumscribed by a single coil, excitation coil, wound coil, or coil. Embodiments of the core tube include a ferromagnetic tube having a uniform tube thickness along its longitudinal axis except for a pair of channels that circumscribe the radial exterior surface of the core tube. Embodiments of the core tube include having a pair of channels that have a thinner cross section of the core tube.

Within the core tube is a cavity that maintains a moveable armature operable to move through the longitudinal axis of the cavity in response to magnetic flux created by current passing through the coil. Embodiments include the core tube being a ferromagnetic tube having a uniform diameter across its longitudinal axis except for a pair of spaced apart channels circumscribing the radial exterior surface of the core tube. Each of the channels are located radially inward from the coil. The two channels are separated from one another along the longitudinal axis by a ferrous spacer. Within each of the channels along the inner radial portion of the channels are a non-magnetic filler material welded to the outer radial surface of the channels. Embodiments include the filler material being replaced by an air gap such that there is an air gap between the radial inner surface of the magnets and the outer radial surface of the core tube. Radially outwardly adjacent the filler material is a magnet or magnets that circumscribe the filler material. Each of the magnets are embedded within the core tube such that the radial exterior surface of the core tube and the radial exterior surface of the magnets are coextensive. A thin wall of the core tube separates the filler material from the inside cavity of the core tube. Embodiments of the magnets are removable from the channels and the core tube such that they can be replaced with new magnets in the instance that the magnets degrade or lose their magnetic force. Embodiments provide that a small gap may exist between the side edges of the magnets and the adjacent core tube surface.

(16) Embodiments of the present disclosure include a solenoid assembly having a core tube that has a Teflon coated sleeve located or disposed along the inner radial surface of the cavity of the core tube. Embodiments provide that the Teflon coated sleeve is operable to reduce friction between the armature and the tube.

(17) It should be appreciated that each magnet that circumscribes the core tube may include multiple curved magnets that are in contact with or close proximity to one another such that in combination they collectively circumscribe the core tube. One embodiment includes that the magnets be C-shaped such that two or more C-shaped magnets are operable to collectively circumscribe the core tube.

(18) In one embodiment, the location of the magnets and the length of the magnets along the longitudinal axis are equal in length to the radially outer most surface of the filler material such that the magnets and the filler material are coextensive in the longitudinal direction. In another embodiment, the magnets are coextensive in the longitudinal direction with the filler material along the edge adjacent the spacer, but are not coextensive on the edge opposite from the spacer such that the magnets extend beyond the filler material. In yet another embodiment, the magnets are coextensive in the longitudinal direction with the filler material along the edge opposite from the spacer, but are not coextensive on the edge adjacent the spacer such that the magnets extend over the spacer. In yet a further embodiment, the magnets are located such that they are radially outward from the filler material. In this embodiment, the magnets are not coextensive with the filler material on either the edge adjacent the spacer or the edge opposite from the spacer. In other words, in this embodiment, the magnets are longer than the filler material in the longitudinal direction and extend past the edge of the filler material that is adjacent the spacer and opposite from the spacer.

(19) Referring to FIG. 1, shown is a cross-sectional view of an exemplary device suitable for use in

practicing exemplary embodiments of this disclosure. Shown in FIG. 1 is a solenoid assembly **100** having a coil **102** (also referred to as an excitation coil or wound coil), a core tube **104**, a first channel **106**, a second channel **108**, and an armature **110**. Core tube **104** includes a longitudinal axis **118**. The coil **102** is enclosed in a housing **112** and located such that the coil **102** is radially outward from the core tube **104**. Coil **102** circumscribes core tube **104**. Coil **102** is operable to have a current pass through it in order to create a magnetic flux that passes through core tube **104** and armature **110** causing armature **110** to move through cavity **115** along the longitudinal axis **118**. Embodiments of coil **102** include it being made of enamel magnet wire made of copper or aluminum, nickel plated magnet wire, ceramic coated magnet wire, extruded jacketed magnet wire such as, but not limited to, extruded materials of TFE, polyether ether ketone (PEEK), polyimide, or any other thin film polyamide-imide, polyester-imide/polyamide-imide coated magnet wire or any combination of materials thereof. Embodiments include coil **102** circumscribing the portion of core tube **104** that includes the first channel **106**, the second channel **108**, and the ferrous spacer **105**. Embodiments of core tube **104** include a cylindrically shaped hollow tube having a radial interior surface **114** and a radial exterior surface **116**. Embodiments of core tube **104** include it being made of a single uniform ferromagnetic material. It should be appreciated that embodiments of core tube **104** include it being made of multiple materials that are welded together. The term ferromagnetic includes ferrous materials and those materials that have a high susceptibility to magnetization. Other than the first channel **106** and the second channel **108**, core tube **104** has a uniform thickness between its radial interior surface **114** and its radial exterior surface **116** throughout its longitudinal axis **118**.

(20) The first channel **106** and the second channel **108** each define a groove disposed on the radial exterior surface **116** of core tube **104**. The first channel **106**, spacer **105** and the second channel **108** are located radially inward from coil **102**. The first channel **106** is spaced apart from the second channel **108** such that they each circumscribe the radial exterior surface **116** of core tube **104** along a set range along the longitudinal axis **118** of core tube **104**. The first channel **106** is spaced apart and separated from the second channel **108** in the longitudinal direction by a ferrous spacer **105**. Ferrous spacer **105** is integral with the core tube **104** and extends radially outward from the radial interior surface **114** of the cavity **115** to the radial exterior surface **114**. Embodiments of ferrous spacer **105** include ferrous spacer **105** having the same uniform thickness as the radial interior surface **114** and radial exterior surface **116** of the remainder of core tube **104** (shown in FIG. 1). Embodiments of ferrous spacer **105** include is having a channel **105a** (shown in FIG. 14) that circumscribes the radial exterior surface of ferrous spacer **105**. Channel **105a** defines a groove disposed on the radial exterior surface of ferrous spacer **105** of core tube **104**. As shown in FIG. 14, channel **105a** is spaced from the first and second channels **106**, **108** such that a portion (**105b**, **105c**) of ferrous spacer **105** has the same uniform thickness as the radial interior surface **114** and radial exterior surface **116** of the remainder of core tube **104**. As illustrated in FIG. 14, channel **105a** does not extend as far radially inward as first and second channels **106**, **108** such that the thickness of core tube **104** radially inward from channel **105** is greater than the thickness of core tube **104** radially inward from the first and second channels **106**, **108**. The first channel **106**, the ferrous spacer **105** and the second channel **108** are located on the core tube **104** such that they are radially outward from the cavity **115**. Embodiments include the radial interior surface **114** having a Teflon coating **119** or a separate tetrafluoroethylene (TFE) sleeve (shown in FIG. 2) that is in contact with the armature **110** outside diameter and the fluid maintained within the cavity **115** is contained within the radial interior surface **114**, and is in contact with the Teflon coating or the TFE inserted sleeve. Embodiments of the TFE sleeve are removably from the cavity **115** of core tube **104** and loose within the cavity **115** of core tube **104** such that it is held or maintained in place within the cavity **115** of core tube **104** be the stopper end **103** of core tube **104**.

(21) The first channel **106** is located such that along its cross-section along the longitudinal axis **118** of core tube **104**, it has a uniform thickness in the radial direction. Similarly, the second

channel **108** is located such that along its cross-section along the longitudinal axis **118** of core tube **104**, it has a uniform thickness in the radial direction. Embodiments of the first channel **106** and the second channel **108** include them maintaining either (i) a filler material **111**, or (ii) an air gap **117** (shown in FIGS. **2** & **14**) along a radially inward portion of each channel. Embodiments of the first channel **106** and the second channel **108** also include them maintaining a first magnetic ring **107** and a second magnetic ring **109**, respectively.

(22) The first magnetic ring **107** is located radially outward from and in contact with either the filler material **111** or the air gap **117** within the first channel **106**. The second magnetic ring **109** is located radially outward from either the filler material **111** or the air gap **117** within the second channel **106**. Embodiments of the first magnetic ring **107** and the second magnetic ring **109** include two or a plurality of curved or C-shaped magnets (shown in FIGS. **4**, **5** and **13**) that circumscribe the core tube **104** within the first and the second channels **106**, **108**, respectively. In the embodiments shown in FIGS. **4**, **5** and **13**, curved or C-shaped magnets are sized such that two or more C-shaped magnets would circumscribe a single core tube **104**. However, it should be appreciated that embodiments of C-shaped magnets include more than two magnets or a plurality of magnets that circumscribe a given core tube **104**. It should be appreciated that embodiments of the first magnetic ring **107** and the second magnetic ring **109** are removably located within the first and second channel **106**, **108**, respectively (shown in FIG. **13**). Embodiments provide that only the magnetic forces from the first magnetic ring **107** and the second magnetic ring **109** maintain the location of the each of the rings within the first and second channels **106**, **108**, respectively.

Embodiments provide that first magnetic ring **107** and the second magnetic ring **109** are not held in place within the first and second channels **106**, **108** by welding or an adhesive between the first and second magnetic rings **107**, **109** and the first and second channels **106**, **108**. In another embodiment, the first and second magnetic ring **107**, **108** are maintained in place within first and second channels **106**, **108** by welding or an adhesive. The first magnetic ring **107** and the second magnetic ring **109** in some embodiments are coextensive with the adjacent filler material or air gap in the longitudinal axis **118** direction (shown in FIGS. **1-3**). As shown in FIGS. **1-3**, the terminal edges **130** of the first magnetic ring **107** and the terminal edges **132** of the second magnetic ring **109** in the longitudinal direction are at the same position along the longitudinal axis **118** as the adjacent filler mater **111** or air gap **117**.

(23) In other embodiments, the first magnetic ring **107** and the second magnetic ring **109** are not coextensive with the adjacent filler material **111** or air gap in the longitudinal axis **118** direction (shown in FIGS. **6-7**). In the embodiments depicted in FIG. **7**, the first magnetic ring **107** is coextensive with the filler material along the edge opposite the second magnetic ring **109**, but extends further in the longitudinal axis **118** direction toward the second magnetic ring **109**.

Similarly, in the embodiment depicted in FIG. **7**, the second magnetic ring **109** is coextensive with the filler material along the edge opposite the first magnetic ring **107**, but extends further in the longitudinal axis **118** direction toward the first magnetic ring **107**.

(24) It should be appreciated that both the first channel **106** and the second channel **108** can include two, three or more surfaces on their radial exterior surface. Referring to FIGS. **8-11**, shown are close-up cross-sectional views of embodiments of the first channel **106** and the second channel **108**. Shown in FIGS. **8-11** are core tube **104** and a portion of the second channel **108**. It should be appreciated that for purposes of this disclosure in FIGS. **8-11**, the cross-section of the first channel **106** is the mirror image of the second channel **108**. Within the second channel **108** is a filler material **111**. However, it should be appreciated that embodiments include second channel **108** having an air gap **117** in place of the filler material **111**. Embodiments of the filler material **111** include non-magnetic material such as, but not limited to, stainless steel, silicon, copper, bronze, aluminum, tin, nickel, epoxy and any combination of the foregoing. Second channel **108** includes a plurality of outer radial surfaces **122**. Embodiments of the plurality of outer radial surfaces **122** can include surfaces that have a straight or flat cross section and surfaces that have a curved cross

section.

(25) Shown in FIG. 8, is second channel **108** with outer radial surfaces **122a**, **122b**, **122c**, and **122d**. As depicted, the cross section of outer radial surface **122a** is flat and extends substantially perpendicular to the longitudinal axis **118** of the core tube **104**. Although not depicted in FIG. 8 (see FIGS. 1-7), outer radial surface **122a** is located adjacent and in contact with the second magnetic ring **109**. The second channel **108** also includes an outer radial surface **122b**, which is adjacent to outer radial surface **122a**. Outer radial surface **122b** has a flat cross section surface and is angled toward outer radial surface **122d**. Outer radial surface **122c** is adjacent to outer radial surface **122b** and is angled such that it is closer to being more perpendicular to the longitudinal axis **118** than outer radial surface **122b**. Outer radial surface **122d** is substantially parallel to the longitudinal axis **118**. It should be appreciated that embodiments of outer radial surface **122d** include it being a shallow tapered surface relative to the longitudinal axis **118**. Outer radial surface **122d** and the interior surface radial surface **114** of core tube **104** define the thin wall **124** of the second channel **108** of core tube **104**.

(26) Reference is now made to the embodiment of second channel **108** illustrated in FIG. 9. Shown in FIG. 9 is second channel **108** having outer radial surfaces **122a**, **122b**, **122c**, and **122d**. Outer radial surface **122a** and **122b** are similar to that shown in FIG. 8. Outer radial surface **122c** in FIG. 9, however, has a convex shaped cross section that is curved outward toward the filler material **120**. Outer radial surface **122c** is adjacent to and between outer radial surfaces **122b** and **122d**. Similar to the embodiment shown in FIG. 8, outer radial surface **122d** is substantially parallel to the longitudinal axis **118**. It should be appreciated that embodiments of outer radial surface **122d** include it being a shallow tapered surface relative to the longitudinal axis **118**. In FIG. 11, outer radial surface **122b** rather than outer radial surface **122c** has a convex shaped cross section that is curved outward toward the filler material **120**. In the embodiment depicted in FIG. 11, outer radial surface **122c** has a flat cross section that is angled relative to the longitudinal axis **118**.

(27) Referring to FIG. 10, shown is a close-up cross-sectional view of another embodiment of second channel **108**. In the embodiment shown in FIG. 10, second channel includes outer radial surfaces **122a**, **122b** and **122d**. Similar to FIGS. 8-9, outer radial surface **122a** is perpendicular to the longitudinal axis **118** and extends between the radial exterior surface **116** and outer radial surface **122b**. In this embodiment, outer radial surface **122d** is substantially parallel to the longitudinal axis **118**. It should be appreciated that embodiments of outer radial surface **122d** include it being a shallow tapered surface relative to the longitudinal axis **118**. Outer radial surface **122b** has a convex shaped cross section that is curved outward toward the filler material **120**. In this embodiment, second channel **108** does not include an outer radial surface **122c**. Outer radial surface **122b** is adjacent to and connects outer radial surface **122a** and **122d**.

(28) Referring to FIG. 12, shown is an exemplary logic flow diagram in accordance with an exemplary method or process of the present disclosure. The exemplary method begins at block **1202**, which states providing a ferromagnetic core tube having a longitudinal axis, a radial exterior surface and a radial interior surface surrounding a cavity, the ferromagnetic core tube comprising a first channel and a second channel disposed on the radial exterior surface of the ferromagnetic core tube, the first channel and the second channel circumscribing the ferromagnetic core tube along a given portion of the longitudinal axis, wherein the first channel and the second channel are spaced from one another along the longitudinal axis, the first channel comprising a plurality of radial exterior surfaces, the second channel comprising a second plurality of radial exterior surfaces, wherein a first portion of the ferromagnetic core tube between the plurality of radial exterior surfaces of the first channel and the radial interior surface of the ferromagnetic core tube define a first thin wall portion of the ferromagnetic core tube, wherein a second portion of the ferromagnetic core tube between the second plurality of radial exterior surfaces of the second channel and the radial interior surface of the ferromagnetic core tube define a second thin wall portion of the ferromagnetic core tube, the first channel comprising a first magnetic ring radially spaced apart

from the first thin wall, the second channel comprising a second magnetic ring radially spaced part from second thin wall; and providing a coil located radially outward of the ferromagnetic core tube, wherein the ferromagnetic core tube comprises a uniform single piece material. Next at block **1204**, it states providing an armature maintained with the cavity, the armature operable to move through the longitudinal axis of the cavity in response to a current passing through the coil.

(29) Next, block **1206** recites wherein the first channel and the second channel are located radially inward from the coil, and wherein the first channel is spaced apart from the second channel along the longitudinal axis by a ferrous spacer. Block **1208** then states wherein the first channel maintains one of a filler material or an air gap between an inner radial surface of the first magnetic ring and the outer radial surface of the first thin wall, and wherein the second channel maintains one of a filler material or an air gap between an inner radial surface of the second magnetic ring and the outer radial surface of the second thin wall. Following block **1208**, block **1210** recites wherein first magnetic ring and the second magnetic ring each include an outer radial surface that is coextensive with the radial exterior surface of the ferromagnetic core tube. Block **1212** then states wherein the first magnetic ring circumscribes the first channel, and wherein the second magnetic ring circumscribes the second channel. Finally, block **1214** states wherein the inner radial surface of the first magnetic ring is one of coextensive or not coextensive along the longitudinal axis with an outer radial surface of the filler material in the first channel, and wherein the inner radial surface of the second magnetic ring is one of coextensive or not coextensive along the longitudinal axis with an outer radial surface of the filler material in the second channel.

(30) Embodiments of the present disclosure include a solenoid assembly having a core tube having a pair of channels that circumscribe the radial exterior surface. The channels include either a filler material or an air gap radially inward from magnetic rings. The solenoid assembly is operable to create a directional magnetic force operable to move an armature located within the core tube through its longitudinal axis. Embodiments include a solenoid having a core tube **104** and a pair of channels **106**, **108** that each have a thin wall section with no welded material. This allows embodiments of the core tube **104** to be able to withstand low internal pressures and the ability to create high magnetic forces. Embodiments of the thin wall section of channels **106**, **108** create a barrier between the fluids within the core tube **104**, which improves the chemical resistance along the radial interior surface of core tube **104**. One of the advantages of having a pair of channels that each have a thin wall portion that is integral with the core tube is to aid in the elimination of leaks without sacrificing the strength of the magnetic forces created by the solenoid.

(31) It is to be understood that any feature described in relation to any one embodiment may be used alone, or in combination with other features described, and may also be used alone, or in combination with one or more features of any other of the embodiments, or any combination of any other of the embodiments. The presently disclosed embodiments are therefore considered in all respects to be illustrative. Furthermore, equivalents and modifications not described above may also be employed without departing from the scope of this disclosure, which is defined in the accompanying claims.

Claims

1. A solenoid assembly comprising: a ferromagnetic core tube having a longitudinal axis, a radial exterior surface and a radial interior surface surrounding a cavity, the ferromagnetic core tube comprising a first channel and a second channel disposed on the radial exterior surface of the ferromagnetic core tube, the first channel and the second channel circumscribing the ferromagnetic core tube along a given portion of the longitudinal axis, wherein the first channel and the second channel are spaced from one another along the longitudinal axis, the first channel comprising a plurality of radial exterior surfaces, the second channel comprising a second plurality of radial exterior surfaces, wherein a first portion of the ferromagnetic core tube between the plurality of

radial exterior surfaces of the first channel and the radial interior surface of the ferromagnetic core tube define a first thin wall portion of the ferromagnetic core tube, wherein a second portion of the ferromagnetic core tube between the second plurality of radial exterior surfaces of the second channel and the radial interior surface of the ferromagnetic core tube define a second thin wall portion of the ferromagnetic core tube, the first channel comprising a first magnetic ring radially spaced apart from the first thin wall, the second channel comprising a second magnetic ring radially spaced apart from second thin wall; and a coil located radially outward of the ferromagnetic core tube, wherein the ferromagnetic core tube comprises a uniform single piece material.

2. The solenoid assembly according to claim 1, the solenoid assembly further comprising an armature maintained with the cavity, the armature operable to move through the longitudinal axis of the cavity in response to a current passing through the coil.

3. The solenoid assembly according to claim 1, wherein the first channel and the second channel are located radially inward from the coil.

4. The solenoid assembly according to claim 1, wherein the first channel is spaced apart from the second channel along the longitudinal axis.

5. The solenoid assembly according to claim 1, wherein the first channel maintains a non-magnetic filler material between an inner radial surface of the first magnetic ring and the outer radial surface of the first thin wall, and wherein the second channel maintains a non-magnetic filler material between an inner radial surface of the second magnetic ring and the outer radial surface of the second thin wall.

6. The solenoid assembly according to claim 5, wherein the inner radial surface of the first magnetic ring is coextensive along the longitudinal axis with an outer radial surface of the filler material in the first channel, and wherein the inner radial surface of the second magnetic ring is coextensive along the longitudinal axis with an outer radial surface of the filler material in the second channel.

7. The solenoid assembly according to claim 5, wherein the inner radial surface of the first magnetic ring is not coextensive along the longitudinal axis with an outer radial surface of the filler material in the first channel, and wherein the inner radial surface of the second magnetic ring is not coextensive along the longitudinal axis with an outer radial surface of the filler material in the second channel.

8. The solenoid assembly according to claim 1, wherein the first channel maintains a first air gap between an inner radial surface of the first magnetic ring and the outer radial surface of the first thin wall, and wherein the second channel maintains a second air gap between an inner radial surface of the second magnetic ring and the outer radial surface of the second thin wall.

9. The solenoid assembly according to claim 1, wherein first magnetic ring and the second magnetic ring each include an outer radial surface that is coextensive with the radial exterior surface of the ferromagnetic core tube.

10. The solenoid assembly according to claim 1, wherein the first magnetic ring circumscribes the first channel, and wherein the second magnetic ring circumscribes the second channel.

11. The solenoid assembly according to claim 1, wherein the ferromagnetic core tube comprises a Teflon coated sleeve located along the radial interior surface surrounding the cavity.

12. The solenoid assembly according to claim 1, wherein the first magnetic ring comprises a first plurality of C-shaped magnets, and wherein the second magnetic ring comprises a second plurality of C-shaped magnets.

13. A solenoid assembly comprising: a ferromagnetic core tube having a longitudinal axis, a radial exterior surface and a radial interior surface surrounding a cavity, the ferromagnetic core tube comprising a first channel and a second channel disposed on the radial exterior surface of the ferromagnetic core tube, the first channel and the second channel circumscribing the ferromagnetic core tube along a given portion of the longitudinal axis, wherein the first channel and the second channel are spaced from one another along the longitudinal axis, the first channel comprising a

plurality of radial exterior surfaces, the second channel comprising a second plurality of radial exterior surfaces, wherein a first portion of the ferromagnetic core tube between the plurality of radial exterior surfaces of the first channel and the radial interior surface of the ferromagnetic core tube define a first thin wall portion of the ferromagnetic core tube, wherein a second portion of the ferromagnetic core tube between the second plurality of radial exterior surfaces of the second channel and the radial interior surface of the ferromagnetic core tube define a second thin wall portion of the ferromagnetic core tube, the first channel comprising a first magnetic ring radially spaced apart from the first thin wall, the second channel comprising a second magnetic ring radially spaced part from second thin wall; an armature maintained with the cavity, the armature operable to move through the longitudinal axis of the cavity in response to a current passing through the coil; a coil located radially outward of the ferromagnetic core tube, wherein the ferromagnetic core tube comprises a uniform single piece material, wherein the first channel maintains a filler material between an inner radial surface of the first magnetic ring and the outer radial surface of the first thin wall, and wherein the second channel maintains a filler material between an inner radial surface of the second magnetic ring and the outer radial surface of the second thin wall.

14. A method comprising: (a) providing a ferromagnetic core tube having a longitudinal axis, a radial exterior surface and a radial interior surface surrounding a cavity, the ferromagnetic core tube comprising a first channel and a second channel disposed on the radial exterior surface of the ferromagnetic core tube, the first channel and the second channel circumscribing the ferromagnetic core tube along a given portion of the longitudinal axis, wherein the first channel and the second channel are spaced from one another along the longitudinal axis, the first channel comprising a plurality of radial exterior surfaces, the second channel comprising a second plurality of radial exterior surfaces, wherein a first portion of the ferromagnetic core tube between the plurality of radial exterior surfaces of the first channel and the radial interior surface of the ferromagnetic core tube define a first thin wall portion of the ferromagnetic core tube, wherein a second portion of the ferromagnetic core tube between the second plurality of radial exterior surfaces of the second channel and the radial interior surface of the ferromagnetic core tube define a second thin wall portion of the ferromagnetic core tube, the first channel comprising a first magnetic ring radially spaced apart from the first thin wall, the second channel comprising a second magnetic ring radially spaced part from second thin wall; and (b) providing a coil located radially outward of the ferromagnetic core tube, wherein the ferromagnetic core tube comprises a uniform single piece material.

15. The method according to claim 14, wherein the first channel and the second channel are located radially inward from the coil, and wherein the first channel is spaced apart from the second channel along the longitudinal axis by a ferrous spacer.

16. The method according to claim 14, wherein the first channel maintains a non-magnetic filler material between an inner radial surface of the first magnetic ring and the outer radial surface of the first thin wall, and wherein the second channel maintains a non-magnetic filler material between an inner radial surface of the second magnetic ring and the outer radial surface of the second thin wall.

17. The method according to claim 14, wherein the first channel maintains a first air gap between an inner radial surface of the first magnetic ring and the outer radial surface of the first thin wall, and wherein the second channel maintains a second air gap between an inner radial surface of the second magnetic ring and the outer radial surface of the second thin wall.

18. The method according to claim 14, wherein the first magnetic ring circumscribes the first channel, and wherein the second magnetic ring circumscribes the second channel.

19. The method according to claim 14, wherein the first magnetic ring comprises a first plurality of curved magnets, and wherein the second magnetic ring comprises a second plurality of curved magnets.
